METRIC
A-A-59441
16 January 2004

COMMERCIAL ITEM DESCRIPTION

VAPOR CORROSION INHIBITORS

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description covers the requirements for vapor corrosion inhibitors (VCIs) to provide corrosion protection for metal components and electrical and electronic assemblies subjected to corrosive environments. The effectiveness of VCIs is governed by the capability to maintain an active molecular layer on the surface of concern.

2. CLASSIFICATION. The VCI materials shall be of the following types:

Type I - VCI-diffusing devices with adhesive backing.

Type II - VCI-diffusing devices without adhesive backing.

Type III - VCI packaged in aerosol containers.

Type IV - Bulk VCI materials in suspension or liquid form.

Type V - Bulk VCI materials in powdered form.

Type VI - VCI packaging products.

Type VII - VCI impregnated wipes.

3. SALIENT CHARACTERISTICS

3.1 Specification sheets. Individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this commercial item description and the specification sheet, the former shall govern.

3.2 Materials. The VCIs shall be formulated from non-toxic compounds, which are environmentally safe. The VCI compounds, other components and adhesive materials shall not support combustion. Aerosol propellants shall be non-flammable and non-toxic. VCI formulations and VCI products shall have passed review by the Naval Environmental Health Center (NEHC). In addition VCI compounds and VCI products shall not contain: a) amines and nitrites or nitrates in combination, b) phosphates, or c) silicones.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

FSC 2090

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3.2.1 Material selection. Care must be taken to ensure that the VCI materials selected are approved for the intended use. In the case of Type VI materials used for parts to be placed in storage aboard ship, the materials selected must meet current requirements for the generation, storage and disposal of waste (residue) at sea.

3.3 Operating environment. The VCI vapor diffusing devices shall retain full functionality for corrosion protection in components and equipment that operate in the temperature range –25 to +65 ºC. The operating environments for each type of VCI are as follows:

3.3.1 Type I. Used in sealed mechanical, electrical and electronic enclosures. Type I VCIs may be used in all equipment: shipboard, land-based, mobile, and aviation. They may also be used in sealed packaging during storage and in transit.

3.3.2 Type II. Used in packaging for protection during storage and in transit. Type II VCIs may be used in all equipment: shipboard, land-based, mobile, and aviation. They may also be used in sealed mechanical, electrical and electronic enclosures when no hazard is created by their movement within the enclosure.

3.3.3 Type III. Used in spot applications in sealed enclosures (medium term protection) and exposed locations (short term or temporary protection). Type III VCIs may be used in all equipment: shipboard, land-based, mobile, and aviation.

3.3.4 Type IV. Used in applications to large components for open storage or transit (medium-term protection). Type IV VCIs are normally used in industrial facilities (intermediate or depot-level).

3.3.5 Type V. Used in applications involving large sealed enclosures-tanks or voids (medium to long term protection). Type V VCIs may be used in shipboard, land-based, and mobile applications.

3.3.6 Type VI. Used in applications involving packaging of materials that must be protected from corrosion. They include long term storage and storage in environments that promote corrosion. Type VI VCIs may be used in all packaging and storage applications.

3.3.7 Type VII. Used in applications involving the performance of maintenance upon equipment, for which short-term protection is required while it is exposed to the environment. Type VII VCIs may be used in all applications.

3.4 Performance. As used in individual spec sheets, VCI materials shall protect classes of metals as defined by the following categories:

MM Multimetal VCI products inhibit corrosion for all metals normally found exposed in electrical and electronic equipment, as well as mechanical/electromechanical equipment and components.

F Ferrous VCI products have been optimized to inhibit corrosion for all ferrous metals found exposed in mechanical, electromechanical, electrical and electronic equipment, including carbon steel, stainless steel, galvanized steel and iron.

NF Non-ferrous products have been optimized to inhibit corrosion for all non-ferrous metals normally found exposed in electrical and electronic equipment, as well as mechanical, electromechanical and metals industries, including copper and its alloys, aluminum and its alloys, and solder.

3.5 Personal safety. The application of VCI materials shall not require the use of personal protective equipment (PPE). Where VCIs are added to other products to improve their performance or application, no additional PPE shall be required then would otherwise be required for the non-VCI formulation of the product.

3.6 Protection life. VCI materials shall provide protection for the periods shown in Table I of specification sheet.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6.1 Shelf life. The shelf life of kits shall be not less than 24 months when stored in a sheltered space having a controlled environment.

6.2 Labeling. The manufacturer's commercial labels for bulk packaging may be used provided that they include the following:

a. Volume or surface area protected.
b. Application conditions.
c. Types of metals for which the product is applicable.
d. Instructions for application or installation.
e. Precautions to be observed during installation.

An expiration date representing the latest installation date for satisfactory performance shall be placed on each kit; it shall read "Install by: [date]."

7. NOTES.

7.1 Intended use. VCI products are coated, impregnated or filled with chemical compounds that actively diffuse invisible corrosion inhibiting vapors into their surrounding atmosphere. The molecules of these vapors adsorb to metal surfaces inside the enclosed volume to form an active layer of protection against corrosion that causes no change to the character of metal surfaces. The VCI layer later completely evaporates from all metal surfaces when they are exposed to the atmosphere and the VCI diffusion source is removed.

7.2 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

PIN code:

AA59441	**/X**	**-**	**X**	**XX**
CID Number	Spec Sheet (see code below)		VCI Type (see code below)	Material Form (see code in Table I of spec sheet)

Spec Sheet	
Sheet	**Code**
1	1
2	2
3	3
4	4
5	5
6	6
7	7

VCI Type	
Type	**Code**
1	A
2	B
3	C
4	D
5	E
6	F
7	G

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Material Form (see also Table I of applicable spec sheet)					
Form	**Code**	**Form**	**Code**	**Form**	**Code**
1	01	11	11	21	21
2	02	12	12	22	22
3	03	13	13	23	23
4	04	14	14	24	24
5	05	15	15	25	25
6	06	16	16	26	26
7	07	17	17	27	27
8	08	18	18	28	28
9	09	19	19	29	29
10	10	20	20	30	30

The example PIN code for a Type IV VCI, material form 18 is: AA59441/4-D18.

7.3 Source of documents.

7.3.1 The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at http://www.arnet.gov/far.

7.3.2 DoD specifications are available online at http://astimage.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.

7.4 National stock numbers (NSNs). See individual specification sheets for NSNs.

7.5 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

c. Quantity required.

7.6 Key words.

Adhesive strip
Aerosol
Corrosion
Diffuser
Diffusing device
Foam
Personal protective equipment
Vapor corrosion inhibitor
VCI

7.7 Definitions.

7.7.1 Enclosure. An enclosure is any cabinet, junction box, termination, or space surrounded by a material layer that provides physical protection to its contents. A “sealed enclosure” is one that completely prevents or restricts the inflow and outflow of air to a small fraction of the total enclosed volume.

7.7.2 Protection life. The period during which the VCI materials will remain active and prevent corrosion when they are installed or applied according to the manufacturer’s instructions.

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7.7.3 Protection volume. For VCI emitters and bulk VCI materials that are applied by weight or volume, the protection volume is the maximum gross volume of the sealed enclosure or closed space for which the individual emitter or recommended application will provide corrosion protection for the rated "protection life." The gross volume is the total volume without a decrement for the included equipment.

MILITARY INTERESTS

Custodians:
Navy – SH
Air Force – 99

Review Activity:
DLA – CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Navy – SH
(Project 2090-0118-000)

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